

NEO Mathematics — Learner Guide

Accuracy, Powers and Scale · Calculating with Standard Form

Driving Question: *Can I calculate with giant and tiny numbers without ever writing out their zeros?*

What are we exploring?

Last lesson you learned to write enormous and tiny numbers in standard form. This lesson those numbers start working: multiplying, dividing, even measuring the distance to the Sun. There is no trick and no machine — just short chains of written working in which every line is equal to the line above it, and every line has a reason you can name.

Your Stepping Stones

- I can multiply and divide numbers in standard form by regrouping the factors and applying the index laws.
- I can rewrite an answer whose coefficient breaks the definition of standard form.
- I can add numbers in standard form by matching the powers first.
- I can say what an answer means in its real context.

Three facts carry the whole lesson

- **Multiplication can be regrouped in any order** — so the ordinary numbers gather together and the powers of ten gather together.
- **The index laws** — multiplying powers adds the exponents; dividing subtracts them (Lesson 03).
- **The definition of standard form** — $a \times 10^n$, where $1 \leq a < 10$ and n is an integer. Your final line must satisfy it.

A worked chain, so you can see the shape

$(3 \times 10^8) \times (5 \times 10^2) = 3 \times 5 \times 10^8 \times 10^2$ (regrouping) $= 15 \times 10^{10}$ ($3 \times 5 = 15$; index law) $= 1.5 \times 10 \times 10^{10}$ ($15 = 1.5 \times 10$) $= 1.5 \times 10^{11}$ (index law again). That last pair of lines is the **rewrite**: 15 broke the definition, so a factor of 10 moved across to the powers. A rewrite line is part of good working, not a mistake.

A possible way through

- **Connect (about 5 minutes):** "Why may the factors change places?" — tap the line that may come next, and meet the definition.
- **Explore (about 20 minutes):** "Growing the working" — choose a story and build the chain line by line. Every choice is a real mathematical decision.
- **Create or Apply (about 12 minutes):** pose your own calculation from the tiles, write its chain in your own hand, then compare with a model chain. Then visit the Practice Companion.
- **Reflect (about 8 minutes):** judge three pieces of handed-in working, rest with the calm four-step route, and notice what this lesson nourished.

The Practice Companion pathway

Each question has support in this order: **Socratic prompt** → **Hint** → **Check** → **Worked solution**. The worked solution unlocks for a question after you have used the prompt, used the hint, and had **two tries**. The support path is always visible.

Using support is part of learning. It is not cheating.

When you feel stuck

- Ask: what is the next *true* line? You only ever need one line at a time.
- Name the fact you are using — regrouping, index law, or the definition. If you cannot name it, pause and look again.
- Open the Scratchpad "Lines of working" page and set the chain out slowly.
- Use a Reconnection Route, then come back. Every route returns you here.

How will I know I am making progress?

- I can write the chain for $(4 \times 10^3) \times (2 \times 10^{\blacksquare})$ without writing out any zeros.
- I can spot that $30 \times 10^{\blacksquare}$ breaks the definition and rewrite it: $30 \times 10^{\blacksquare} = 3 \times 10 \times 10^{\blacksquare} = 3 \times 10^{\blacksquare}$.
- I can explain why $3 \times 10^{\blacksquare} + 2 \times 10^{\blacksquare}$ is not $5 \times 10^{\blacksquare}$.

Before you finish

Your best chain — especially one with the rewrite line — deserves a place in the Cosmic Fact File. One idea worth carrying forward is enough.

You can pause. You can return. You do not need to complete everything in one sitting.

NEO by Nudge Education · Foundations pathway · Connection before curriculum. Always.